

API Specification

Project: LearnLens AI

Protocol: REST (HTTP/1.1) & WebSockets (RFC 6455)

Base URL: `http://127.0.0.1:8000`

1. System & Diagnostic Endpoints

`GET /api/system/status`

Returns runtime connectivity status for Ollama, faster-whisper, and OCR engines.

- Response: `200 OK`

```
{
  "ollama_connected": true,
  "ollama_models": ["qwen2.5:latest", "qwen2.5-vl:latest"],
  "whisper_available": true,
  "ocr_available": true,
  "active_session_id": null,
  "current_task_state": "IDLE",
  "system_load": {"cpu": "Normal", "memory": "Available"}
}
```

`POST /api/demo/seed`

Seeds a verified educational demo session ("Cybersecurity & Cloud Fundamentals - Module 2: Network Firewalls") with pre-rendered architecture slides and transcripts.

- Response: `200 OK`

```
{
  "status": "SUCCESS",
  "session_id": "demo_cybersecurity_module_2",
  "message": "High-yield educational demo session loaded!"
}
```

2. Session Management Endpoints

`POST /api/sessions`

Initializes a new learning session namespace.

- Request Body:

```
{
  "title": "Coursera - Deep Learning Week 1",
  "source_platform": "Coursera",
  "source_url_or_title": "Neural Networks Overview"
}
```

- Response: `200 OK` (Returns full `LearningSession` schema)

`GET /api/sessions`

Lists all active and completed sessions sorted by `updated\_at DESC`.

`DELETE /api/sessions/{session\_id}`

Cascading purge of session database records, captured frames, and generated artifacts.

3. Conversational AI Teacher Endpoints

`POST /api/sessions/{session\_id}/chat`

Sends a student question to the grounded AI Teacher.

- Request Body:

```
{
  "session_id": "demo_cybersecurity_module_2",
  "message": "What is a stateful firewall?",
  "mode": "simple",
  "cross_session": false
}
```

- Response: `200 OK`

```
{
  "id": "msg_9812",
  "session_id": "demo_cybersecurity_module_2",
  "sender": "assistant",
  "text": "Based on your captured lesson at 14:32...",
  "mode": "simple",
  "timestamp": 1789719662.0,
  "evidence": [
    {
      "timestamp": "14:32",
      "frame_id": "frame_demo_02",
      "thumbnail_url": "/api/frames/frame_demo_02/thumbnail",
      "quote": "A stateful firewall tracks the entire lifecycle of a TCP handshake.",
      "concept_name": "Stateful Inspection"
    }
  ]
}
```

4. Human Control & Interruption Endpoints

`POST /api/sessions/{session\_id}/pause`

Transitions task agent state to `PAUSED`. Suspends frame ingestion.

`POST /api/sessions/{session\_id}/resume`

Transitions task agent state back to `OBSERVING`.

`POST /api/sessions/{session\_id}/command`

Executes natural language commands.

- Request Body: ``{"session_id": "...", "command": "Generate PPT"}``

5. Artifact Generation Endpoints

`POST /api/sessions/{session\_id}/generate-ppt`

Builds a 16:9 widescreen PowerPoint presentation.

- **Request Body:** `{ "session\_id": "...", "style": "teaching", "slide\_count": 8 }`

- **Response:** `{ "status": "SUCCESS", "filename": "LearnLens\_...pptx", "download\_url": "/api/exports/..." }`

`POST /api/sessions/{session\_id}/generate-pdf`

Builds PDF A (Visual Study Pack) or PDF B (AI Teaching Report).

- **Request Body:** `{ "session\_id": "...", "pdf\_type": "teaching\_report" }`

6. WebSocket Protocol

`ws://127.0.0.1:8000/ws/session/{session\_id}`

- **Client -> Server Events:**

- `transcript\_chunk`: `{ "type": "transcript\_chunk", "text": "...", "timestamp\_sec": 14.2 }`

- `frame\_capture`: `{ "type": "frame\_capture", "image\_base64": "data:image/jpeg;base64,...", "timestamp\_sec": 14.2 }`

- **Server -> Client Events:**

- `transcript\_received`: Broadcasts segment and newly extracted concepts.

- `frame\_analyzed`: Broadcasts deduplicated frame metadata and OCR.